SQL SERVER

Always On Availability Groups

100+ Real-Time FAQs — Pointwise Answers

Production-focused interview & operations reference

On-Premises • AWS EC2 • Azure VM • Hybrid

PMSQLDBA • sqldbachamps.com

Senior SQL Server DBA Reference Series

Table of Contents

Table of Contents 2
About This Document 2
1. Fundamentals & Architecture 3
2. Availability Modes & Failover 5
3. Configuration & Setup 7
4. Listener & Client Connectivity 10
5. Readable Secondaries & Read-Only Routing 12
6. Backup & Recovery 14
7. Monitoring & Health 16
8. Real-World Troubleshooting 18
9. Performance & Data Synchronization 21
10. Patching, Maintenance & Upgrades 23
11. Cloud & Hybrid Deployments 25
12. Advanced Topics & Best Practices 27

About This Document

This reference compiles 100+ real-time, scenario-driven Q&As on SQL Server Always On Availability Groups, with every answer laid out as concise bullet points for fast scanning and recall. It is organized into twelve practical domains — from fundamentals and configuration through live troubleshooting, performance, patching, and cloud/hybrid deployment.

Use it as interview preparation, an on-call quick reference, or a knowledge-base seed. Answers reflect behavior on SQL Server 2016 through 2022 across on-premises, AWS EC2, and Azure VM environments.

> **TIP** Always route application connections through the AG listener with MultiSubnetFailover=True and connection-retry logic — direct server-name connections defeat the point of an AG.

1. Fundamentals & Architecture

Q1. What is an Always On Availability Group (AG)?

- A high-availability + disaster-recovery feature that protects a set of user databases as a single unit.
- One primary replica holds the read-write copy; up to eight secondary replicas hold synchronized copies.
- Runs on a Windows Server Failover Cluster (WSFC) but needs no shared storage — each replica has its own local copy.

Q2. Difference between an AlwaysOn Failover Cluster Instance (FCI) and an Availability Group?

- FCI protects an entire instance using shared storage; only one node owns the storage at a time (single copy of data).
- AG protects individual databases and keeps multiple independent local copies (readable secondaries + DR).
- They can be combined — an FCI can act as a replica inside an AG.

Q3. What is WSFC and why is it required?

- Windows Server Failover Clustering provides health monitoring, quorum, and the virtual name/IP for the listener.
- Traditional AGs register as a clustered resource so WSFC can detect failures and coordinate automatic failover.
- SQL 2017+ also allows a clusterless read-scale AG (CLUSTER_TYPE=NONE) — but with no automatic failover.

Q4. What is quorum and which quorum models exist?

- Quorum is the voting mechanism WSFC uses to decide if enough healthy members exist to keep running (avoids split-brain).
- Models: Node Majority, Node & File Share Majority, Node & Disk Majority, and Cloud Witness (Azure blob).
- Best practice: keep an odd number of votes — typically nodes + a witness.

Q5. How many replicas can an AG have?

- Enterprise Edition: up to 9 total (1 primary + 8 secondaries).
- Up to 3 replicas can be synchronous-commit.
- Up to 2 replicas can be automatic-failover partners.

Q6. Difference between a primary and a secondary replica?

- Primary hosts the read-write databases and ships transaction log records to secondaries.
- Secondaries receive and harden/redo those log records.
- Secondaries can optionally serve read-only workloads and backups; one is promoted to primary on failover.

Q7. Prerequisites for a database to join an AG?

- Must be a user database (not system), online, read-write, and multi-user.
- Must be in FULL recovery model with at least one full backup taken.
- Cannot already be in another AG and must not use AUTO_CLOSE.
- FILESTREAM and contained-DB settings must be consistent across replicas.

Q8. How does an AG differ from Database Mirroring?

- Mirroring is deprecated: one database, two servers (+ optional witness), a single non-readable mirror.
- AGs support multiple databases as a group and up to 8 secondaries.
- AGs add readable secondaries, backups on secondaries, a listener, and richer failover policies.

Q9. Which SQL Server editions support AGs?

- Enterprise Edition — full AGs (multiple secondaries, readable secondaries).
- Standard Edition — only Basic Availability Groups.

Q10. What is a Basic Availability Group?

- A Standard Edition feature offering minimal HA.
- Exactly two replicas and a single database per AG.
- No read access or backups on the secondary; cannot add replicas to a running Basic AG.
- Effectively replaces deprecated Database Mirroring on Standard Edition.

2. Availability Modes & Failover

Q11. Synchronous-commit vs asynchronous-commit mode?

- Sync-commit: primary waits for the secondary to harden the log before committing — zero data loss, added latency.
- Async-commit: primary commits without waiting — no latency cost, but possible data loss on failover.
- Async is typical for distant/high-latency DR replicas.

Q12. Requirements for automatic failover?

- Both primary and target secondary in synchronous-commit mode.
- Both set to automatic failover and currently in the SYNCHRONIZED state.
- WSFC must have quorum. If any condition fails, only manual/forced failover is possible.

Q13. What is a manual (planned) failover?

- Administrator-initiated failover to a SYNCHRONIZED sync-commit secondary with no data loss.
- Used for planned maintenance.
- Graceful role swap — the old primary becomes a secondary.

Q14. What is a forced failover and when is it used?

- Promotes a secondary that may not be fully synchronized (FORCE_FAILOVER_ALLOW_DATA_LOSS).
- Used in disaster scenarios (e.g., an async DR replica).
- Can lose committed transactions; the former primary must be resumed manually and may need reseeding.

```
ALTER AVAILABILITY GROUP [AG_Prod]
  FORCE_FAILOVER_ALLOW_DATA_LOSS;
```

Q15. What is the flexible failover policy / failover condition level?

- Controls how sensitive automatic failover is, via FAILURE_CONDITION_LEVEL (1–5) and HEALTH_CHECK_TIMEOUT.
- Level 1 = trigger only on server-down; higher levels also react to unresponsive service, critical errors, exhausted worker threads.
- Default is level 3.

Q16. What are the lease timeout and health-check timeout?

- Health-check timeout (default 30s): how long WSFC waits for sp_server_diagnostics before flagging the replica unhealthy.
- Lease timeout (default 20s): heartbeat between the SQL resource DLL and the instance to prevent split-brain.
- Tune carefully — aggressive values cause false failovers.

Q17. Difference between planned and unplanned failover?

- Planned: admin-initiated on a healthy, synchronized system — no data loss.
- Unplanned: triggered automatically by WSFC on a failure.
- Unplanned is only lossless if the target was SYNCHRONIZED at the moment of failure.

Q18. What happens to in-flight (uncommitted) transactions during failover?

- Uncommitted transactions are rolled back during recovery of the new primary.
- Applications see a brief outage and must reconnect (via the listener) and retry in-flight work.
- Committed transactions already hardened on the target are preserved.

Q19. How many automatic-failover targets can you configure?

- Up to two replicas can be automatic-failover partners (SQL 2016+).
- This allows chained failover if the primary and its first partner both fail.

Q20. Can you fail over a single database within an AG?

- No — failover happens at the AG level; all databases move together.
- For per-database failover, put those databases in separate AGs.
- With DB_FAILOVER=ON (SQL 2016+), a single database failure can trigger a whole-group failover.

3. Configuration & Setup

Q21. High-level prerequisites to configure an AG?

- All nodes in the same WSFC and same AD domain.
- AlwaysOn enabled on each SQL instance; consistent versions/collations/patch levels.
- Databases in FULL recovery with a full backup taken.
- Firewall open for the mirroring endpoint (default 5022) and the listener; matching drive paths recommended.

Q22. How do you enable the Always On feature?

- Open SQL Server Configuration Manager > SQL Server service properties.
- On the AlwaysOn High Availability tab, tick 'Enable AlwaysOn Availability Groups' (node must be in a WSFC).
- Restart the SQL Server service; repeat on every replica.

Q23. What is a database mirroring endpoint and why is it needed?

- The TCP endpoint (default port 5022) over which replicas exchange log data.
- Each instance in an AG needs exactly one such endpoint.
- Partner service accounts must have CONNECT permission on each other's endpoints.

```
CREATE ENDPOINT [Hadr_endpoint]
  STATE = STARTED
  AS TCP (LISTENER_PORT = 5022)
  FOR DATA_MIRRORING (ROLE = ALL,
    ENCRYPTION = REQUIRED ALGORITHM AES);
```

Q24. Automatic seeding vs backup/restore seeding?

- Automatic seeding (SEEDING_MODE=AUTOMATIC, SQL 2016+) streams the DB directly over the endpoint — no manual backup/restore.
- Needs matching file paths; enable trace flag 9567 for stream compression.
- Backup/restore (manual) seeding gives more control for very large DBs over slow links.

Q25. Which network ports must be open?

- Mirroring endpoint port (default 5022) between all replicas.
- SQL Server port (1433 or custom) for client + listener traffic.
- WSFC/cluster ports (e.g., 3343 plus dynamic RPC); cloud also needs the load balancer probe port.

Q26. Service-account considerations for AGs?

- Domain accounts get endpoint permissions granted automatically.
- Different/built-in accounts require creating logins for partner service accounts + GRANT CONNECT ON ENDPOINT.
- A common domain account or gMSA greatly simplifies setup.

Q27. How do you add a new database to an existing AG?

- Ensure FULL recovery with a fresh full backup.
- Add it on the primary, then join/seed on each secondary.
- With automatic seeding, grant the AG CREATE ANY DATABASE on secondaries first.

```
-- On secondary (once per AG):
ALTER AVAILABILITY GROUP [AG_Prod]
  GRANT CREATE ANY DATABASE;
-- On primary:
ALTER AVAILABILITY GROUP [AG_Prod]
  ADD DATABASE [SalesDB];
```

Q28. Why must AG databases be in FULL recovery model?

- AGs synchronize by shipping transaction log records.
- SIMPLE / BULK_LOGGED truncate or minimally log, breaking the continuous log chain.
- FULL recovery guarantees every change is fully logged.

Q29. How do you configure a multi-subnet AG?

- Place replicas in different subnets and add one listener IP per subnet.
- Set RegisterAllProvidersIP=1 so DNS registers all listener IPs.
- Clients connect with MultiSubnetFailover=True for fast failover.

Q30. Core T-SQL steps to create an AG?

- Create endpoints on all replicas.
- Create the AG on the primary, specifying replicas and options.
- Join and seed on each secondary (SSMS wizards generate this; scripting keeps it repeatable).

```
CREATE AVAILABILITY GROUP [AG_Prod]
  FOR DATABASE [SalesDB]
  REPLICA ON
   N'SQLP1' WITH (ENDPOINT_URL='TCP://SQLP1:5022',
     AVAILABILITY_MODE=SYNCHRONOUS_COMMIT,
     FAILOVER_MODE=AUTOMATIC,
     SEEDING_MODE=AUTOMATIC),
   N'SQLP2' WITH (ENDPOINT_URL='TCP://SQLP2:5022',
     AVAILABILITY_MODE=SYNCHRONOUS_COMMIT,
     FAILOVER_MODE=AUTOMATIC,
     SEEDING_MODE=AUTOMATIC);
```

4. Listener & Client Connectivity

Q31. What is an Availability Group listener?

- A virtual network name (VNN) plus one or more virtual IPs clients connect to instead of a specific server.
- WSFC keeps the listener pointed at the current primary.
- Failover becomes transparent — apps keep the same connection string.

Q32. How does the listener route connections?

- By default it directs connections to the current primary.
- With ApplicationIntent=ReadOnly and read-only routing configured, it redirects to a readable secondary.
- The routing list controls which secondary is used.

Q33. What does MultiSubnetFailover=True do?

- Tells the driver to try all listener IPs simultaneously instead of sequentially.
- Cuts multi-subnet failover connection time from ~20+ seconds to sub-second.

Q34. Why do connections fail over slowly in multi-subnet without the flag?

- The client tries listener IPs one at a time, waiting for TCP timeouts on the offline subnet's IP.
- Lowering HostRecordTTL and RegisterAllProvidersIP=1 help legacy clients.
- The real fix is MultiSubnetFailover=True.

Q35. Can an AG have more than one listener?

- SSMS supports one listener per AG.
- Additional listeners can be added via PowerShell/WSFC (extra client access points + network-name resources).
- Used to separate read-write vs read-only names.

Q36. What AD/DNS objects does creating a listener produce?

- A WSFC network-name resource (a Virtual Computer Object / VCO in Active Directory).
- DNS A records for each listener IP.
- The CNO needs 'Create Computer objects' rights in the OU, or the VCO must be pre-staged.

Q37. What are RegisterAllProvidersIP and HostRecordTTL?

- RegisterAllProvidersIP (default 1 for multi-subnet): DNS returns all listener IPs so clients try them in parallel.
- HostRecordTTL (default 1200s): how long clients cache the DNS record.
- Lower HostRecordTTL (60–300s) speeds re-resolution for legacy clients.

Q38. How do you test listener connectivity?

- nslookup the listener name to confirm it resolves to all IPs.
- Connect via sqlcmd/SSMS and query @@SERVERNAME to see which replica answered.
- Re-test after a manual failover to confirm the listener follows the primary.

```
sqlcmd -S AGListener,1433 -E -Q "SELECT @@SERVERNAME"
```

5. Readable Secondaries & Read-Only Routing

Q39. What is a readable secondary?

- A secondary configured to allow read-only connections (ALLOW_CONNECTIONS = READ_ONLY or ALL).
- Offloads reporting/read queries from the primary.
- Continuously applies redo from the log stream while serving reads.

Q40. What is read-only routing and how is it configured?

- Automatically redirects read-intent connections (via the listener) to readable secondaries.
- Set a READ_ONLY_ROUTING_URL on each readable replica.
- Set a READ_ONLY_ROUTING_LIST on the primary to order/load-balance secondaries.

```
ALTER AVAILABILITY GROUP [AG_Prod]
 MODIFY REPLICA ON N'SQLP1' WITH
 (PRIMARY_ROLE (READ_ONLY_ROUTING_LIST=
   (('SQLP2','SQLP3'),'SQLP2')));
```

Q41. What is a read-scale (clusterless) AG?

- An AG created with CLUSTER_TYPE=NONE (SQL 2017+) for read-scale-out without a WSFC.
- No automatic failover or HA — only manual failover and read offloading.
- Works across Windows/Linux.

Q42. How do read workloads on a secondary affect the database?

- Readable secondaries force row-versioning (snapshot isolation), adding 14 bytes per row over time.
- The version store consumes tempdb.
- Long-running reads can delay ghost-record cleanup and, in some cases, redo progress.

Q43. What does ApplicationIntent=ReadOnly do?

- A connection-string keyword that flags the connection as read-only.
- Combined with read-only routing, it redirects the connection to a readable secondary.

Q44. Can you load-balance read-only connections across multiple secondaries?

- Yes (SQL 2016+) by grouping replicas in nested parentheses in the routing list.
- Connections are distributed round-robin within a group before moving to the next.

Q45. Can you take backups on a readable secondary?

- Yes — copy-only full backups and transaction-log backups are supported.
- Differential and non-copy-only full backups are not supported on secondaries.
- Backup preference settings control which replica the jobs target.

6. Backup & Recovery

Q46. On which replicas can you take backups in an AG?

- Full copy-only and log backups: primary or any secondary.
- Regular (non-copy-only) full and differential backups: primary only.
- The automated backup preference tells your jobs where to run.

Q47. What backup preference options exist for an AG?

- Prefer Secondary (default), Secondary only, Primary, and Any Replica.
- Each replica has a backup priority (0–100).
- Jobs should call sys.fn_hadr_backup_is_preferred_replica to decide whether to run locally.

Q48. How do transaction-log backups work across replicas?

- There is a single shared log chain across all replicas.
- A log backup on any replica truncates the log for the whole AG.
- Point all log-backup jobs at the preferred replica to keep the chain consistent.

Q49. How do you make Agent backup jobs AG-aware?

- Wrap backup logic so it runs only where sys.fn_hadr_backup_is_preferred_replica returns 1.
- Deploy the same job on every replica.
- Ola Hallengren's scripts and AG-aware maintenance plans handle this automatically.

```
IF sys.fn_hadr_backup_is_preferred_replica('SalesDB') = 1
BEGIN
  BACKUP LOG [SalesDB] TO DISK = N'\\backup\SalesDB.trn';
END
```

Q50. How do you restore a database that is part of an AG?

- You cannot restore over a database while it is in the AG.
- Remove the DB from the AG (on the replica or whole AG).
- Restore on the primary, then re-add and re-seed to secondaries.

Q51. Why must full backups on a secondary be copy-only?

- A normal full backup resets the differential base and disrupts the shared backup/log chain.
- Copy-only backups don't affect the base or log chain — safe to run anywhere.

Q52. The transaction log on the primary keeps growing — AG-specific causes?

- A secondary not synchronizing or with data movement suspended (log_reuse_wait_desc = AVAILABILITY_REPLICA).
- The log can't truncate past a point a secondary still needs.
- Also check a stalled log-backup job and a large redo/log-send queue.

7. Monitoring & Health

Q53. Which DMVs are essential for monitoring an AG?

- sys.dm_hadr_availability_replica_states — roles and sync health.
- sys.dm_hadr_database_replica_states — send/redo queues, LSNs, last_commit_time.
- sys.dm_hadr_availability_group_states, sys.availability_groups, sys.availability_replicas.
- sys.dm_hadr_availability_replica_cluster_nodes for the WSFC view.

Q54. What is the Always On Dashboard?

- An SSMS dashboard (right-click AG > Show Dashboard).
- Visualizes roles, sync state, failover readiness, queue sizes, and estimated data loss.
- The fastest first stop for a health check.

Q55. What do the synchronization states mean?

- SYNCHRONIZED — sync-commit secondary fully caught up and failover-ready.
- SYNCHRONIZING — receiving data but not guaranteed lossless (async, or a sync replica catching up).
- NOT SYNCHRONIZING — data movement is broken/suspended and the replica is falling behind.

Q56. What are the send queue and redo queue?

- Log-send queue (KB): log on the primary not yet sent to the secondary — drives data loss / RPO.
- Redo queue (KB): log received on the secondary not yet replayed — drives failover time / RTO.

Q57. How do you estimate potential data loss (RPO) for an async replica?

- Compare last hardened LSN/last_commit_time on the secondary vs the primary.
- The gap approximates data at risk.
- A large, growing log-send queue signals increasing potential loss.

Q58. How do you estimate failover/recovery time (RTO)?

- Driven mainly by redo queue size and the secondary's redo rate (redo_queue_size / redo_rate).
- Add database recovery time.
- Monitor redo_queue_size and redo_rate in sys.dm_hadr_database_replica_states.

Q59. Which Extended Events / logs help diagnose AG issues?

- The AlwaysOn_health XE session (auto-created) captures state changes, errors, and failovers.
- Combine with the SQL error log and Windows System event log.
- Use the WSFC cluster log (Get-ClusterLog) for the cluster-side timeline.

Q60. What do is_suspended and suspend_reason indicate?

- is_suspended=1 means data movement for that DB on that replica is paused.
- suspend_reason explains why (user action, redo error, capacity, etc.).
- Suspended movement stops catch-up and blocks log truncation on the primary.

8. Real-World Troubleshooting

Q61. Secondary shows 'Not Synchronizing / Recovery Pending' — what do you check?

- Confirm the DB is online and the endpoint is started and reachable.
- Check the error log for redo errors, out-of-disk, or a broken log chain.
- Verify space for the version store and data/log files.
- Resume data movement for transient cases; reseed if the log chain is broken.

Q62. Redo queue on a secondary keeps building — likely causes?

- Slow secondary storage or CPU pressure.
- Single-threaded redo blocked by long-running read queries holding schema locks.
- Very high change rate on the primary.
- Check redo_rate, storage latency, and blocking; parallel redo (SQL 2016+) helps.

Q63. Data movement is suspended on one database — how to resume?

- Find and fix the root cause in the error log first.
- Then resume movement for that database.
- If the log chain is broken, remove the DB from the AG on that secondary and reseed.

```
ALTER DATABASE [SalesDB] SET HADR RESUME;
```

Q64. The AG failed over unexpectedly overnight — how do you root-cause it?

- Build a timeline from AlwaysOn_health XE, the SQL error log, and Get-ClusterLog.
- Look for lease/health-check timeouts, a service hang, a node reboot, or network partition.
- Correlate with Windows event logs and any AV/backup activity.

Q65. The listener is unreachable after a failover — what do you investigate?

- Confirm the listener network-name and IP cluster resources are online on the new owner.
- Check DNS resolves to the correct IP(s) and the VCO isn't disabled in AD.
- Verify firewall rules; in cloud, check the load balancer backend pool and health probe.

Q66. Apps get 'The connection to the primary replica is not active' — meaning?

- The secondary can't reach the primary over the mirroring endpoint (error 35250).
- Check endpoint state (STARTED) and the 5022 port/firewall.

- Verify service-account CONNECT permission on endpoints and network connectivity.

Q67. A database is stuck 'In Recovery' or 'Reverting' on a secondary — meaning?

- The replica is replaying/undoing log to reach a consistent state.
- Common after a forced failover or resume; large redo backlogs make Reverting slow.
- Monitor via the error log and percent-complete DMVs; don't interrupt it.

Q68. The log-send queue is growing on a DR replica — how to diagnose?

- Usually the network can't keep up with the primary's log-generation rate.
- Measure bandwidth/latency; check for saturation or packet loss.
- Consider log-stream compression and reduce bulk operations during peak.

Q69. Automatic seeding fails or hangs — common reasons?

- Mismatched file paths on the secondary or insufficient disk space.
- AG lacks CREATE ANY DATABASE on the secondary.
- Endpoint/firewall issues, or a very large DB timing out over a slow link.
- Check sys.dm_hadr_automatic_seeding and sys.dm_hadr_physical_seeding_stats.

Q70. A WSFC node goes down — what happens to the AG?

- If it hosted the primary and a sync auto-failover partner is SYNCHRONIZED with quorum, the AG fails over automatically.
- If it was a secondary, the AG stays up but loses that copy's protection/readability.
- Loss of quorum stops the AG entirely.

Q71. How do you prevent and recover from a split-brain scenario?

- Prevent with correct quorum/witness (odd votes) and the lease-timeout mechanism.
- If it occurs after a forced failover, pick the authoritative primary.
- Force it online and reseed the other replicas from it to discard divergent data.

Q72. Error 35250 'connection to the primary is not active' repeatedly — fix?

- Verify mirroring endpoints are STARTED on all replicas.
- Confirm port 5022 is open both ways and encryption algorithms match.
- Ensure service accounts have CONNECT on partner endpoints; recreate/restart the endpoint if stopped.

Q73. The cluster loses quorum — impact and recovery?

- WSFC shuts down cluster services and the AG becomes unavailable (DBs go resolving/not-online).
- Recover by bringing enough nodes/witness back online.
- Or force-start on a surviving node (Start-ClusterNode -ForceQuorum), then reassess votes.

Q74. Fix data movement suspended on just one DB while others are healthy?

- It's per-database — treat it individually.
- Read suspend_reason, fix the issue (space, redo error), then SET HADR RESUME for that DB only.
- If the log chain is broken, remove and reseed just that database.

9. Performance & Data Synchronization

Q75. Common causes of synchronization latency?

- Network latency/bandwidth between replicas.
- Slow secondary storage (redo) or CPU pressure.
- Large/long-running transactions on the primary and blocking from read workloads.
- Sync-commit mode adds inherent commit round-trip latency.

Q76. What is the HADR_SYNC_COMMIT wait type?

- The wait on the primary while a commit blocks until a sync secondary hardens the log.
- High values usually mean network latency or slow log-flush on the secondary.
- A direct performance cost of sync-commit.

Q77. How does synchronous-commit affect primary performance?

- Every commit adds a network round-trip to the sync secondary.
- Latency is proportional to inter-replica latency.
- Keep sync replicas close (LAN/AZ); use async for distant DR replicas.

Q78. How can you reduce redo latency on a secondary?

- Use fast, low-latency storage and adequate CPU.
- Minimize long-running read queries that block redo.
- Keep the secondary at the same or better spec; rely on parallel redo (SQL 2016+).

Q79. What is parallel redo?

- SQL 2016+ uses multiple redo threads per database on secondaries (bounded by CPUs).
- Improves redo throughput and reduces failover time vs single-threaded redo.

Q80. How do you size network bandwidth for an AG?

- Baseline the primary's peak log-generation rate (log backups / Log Bytes Flushed/sec).
- The link must sustain that rate plus headroom, or the send queue grows.
- Account for compression and multiple secondaries multiplying traffic.

Q81. How do large transactions impact an AG?

- A single huge transaction sends a big log burst that must be shipped and redone.
- It can spike send/redo queues, delay log truncation, and lengthen failover recovery.
- Prefer batching large operations into smaller committed chunks.

10. Patching, Maintenance & Upgrades

Q82. Correct order to patch/upgrade a synchronous AG with minimal downtime?

- Patch secondaries first, one at a time; validate each.
- Fail over from primary to a patched, SYNCHRONIZED secondary (brief outage).
- Patch the old primary; optionally fail back.

Q83. How do you patch with minimal downtime in auto-failover setups?

- Temporarily set the primary's failover mode to manual to avoid an unintended auto-failover.
- Patch secondaries, then do a planned manual failover to a patched secondary.
- Patch the former primary and restore automatic failover afterwards.

Q84. How do you upgrade the SQL Server major version across an AG?

- Use a rolling upgrade: upgrade secondaries first (they can be a higher version temporarily).
- Fail over to an upgraded secondary, then upgrade the old primary.
- Do not fail back to a lower version once the DB format is upgraded.

Q85. Can you add or remove a replica while the AG is online?

- Yes — adding a replica (with automatic seeding) and removing a secondary are online operations.
- Adding a sync auto-failover partner is also online.
- Re-validate quorum votes afterward.

Q86. Index-maintenance considerations on AGs?

- Rebuilds generate large log volumes that must sync and redo — spiking queues.
- Prefer ONLINE rebuilds and off-peak windows.
- Consider targeted REORGANIZE or resumable index operations to limit log bursts.

Q87. How do you remove a database from an AG?

- Remove it on the primary — copies on secondaries go to a restoring state.
- Drop the leftover copies on each secondary to fully retire it.

```
-- On primary:
ALTER AVAILABILITY GROUP [AG_Prod]
  REMOVE DATABASE [SalesDB];
```

```
-- On each secondary (optional cleanup):
DROP DATABASE [SalesDB];
```

Q88. How do you migrate an AG to new hardware?

- Add the new servers as WSFC nodes and as secondaries; seed them.
- Let them synchronize, then fail over to a new node.
- Remove the old nodes/replicas — a near-zero-downtime migration.

11. Cloud & Hybrid Deployments

Q89. Key considerations for running an AG on Azure VMs?

- Use an Availability Set or Availability Zones for the VMs.
- Use an Azure Internal Load Balancer (ILB) for the listener (no gratuitous ARP in Azure).
- Use a Cloud Witness for quorum and premium/ultra disks for log/redo.
- Enable accelerated networking to reduce sync latency.

Q90. Why does an AG listener on Azure need a load balancer?

- Azure doesn't honor the floating VIP/ARP behavior on-prem listeners rely on.
- An ILB with a health probe and floating IP (DSR) routes traffic to the current listener owner.

Q91. How do you configure an AG listener on AWS EC2?

- Common patterns: a Network Load Balancer fronting the replicas (health check to the primary).
- Or assign multiple secondary private IPs (one per subnet/AZ) with RegisterAllProvidersIP.
- Multi-AZ deployments need a witness reachable from all nodes.

Q92. What is a Distributed Availability Group and when do you use it?

- It spans two separate AGs, each on its own WSFC (often different data centers/clouds).
- Ideal for cross-region/hybrid DR and migrations.
- The two clusters are loosely coupled and communicate via a forwarder — no shared WSFC.

Q93. How do you provide a quorum witness in the cloud?

- Use a Cloud Witness (Azure Storage blob) — cheap and region-independent (WS 2016+).
- Or a File Share Witness on a highly available share.
- The witness supplies the tie-breaking vote for an even number of nodes.

Q94. How does an AG differ from an Azure shared disk / FCI in the cloud?

- FCI needs shared storage (Azure shared disks / Premium Files) and gives one instance-level copy.
- AG uses independent local disks per VM with multiple readable copies + cross-region DR.
- AGs are generally preferred in cloud (no shared-storage constraints).

Q95. What latency considerations apply to multi-region cloud AGs?

- Cross-region latency makes sync-commit impractical (it throttles every commit).
- Use async-commit for distant regions with accepted RPO.
- Keep sync auto-failover replicas within a region/zone; use async or a Distributed AG for remote DR.

12. Advanced Topics & Best Practices

Q96. What is a Contained Availability Group (SQL Server 2022)?

- Stores instance-level metadata — logins, Agent jobs, linked servers — in contained system DBs inside the AG.
- Logins and jobs automatically fail over with the AG.
- Eliminates much of the manual sync work between replicas.

Q97. How do you keep SQL Agent jobs consistent across replicas?

- Deploy identical jobs on all replicas.
- Add a guard step checking the local role (sys.fn_hadr_is_primary_replica) so work runs only on the primary.
- SQL 2022 contained AGs solve this by containing the jobs in the AG.

Q98. How do you handle logins and orphaned users on secondaries?

- SQL logins created only on the primary aren't in the secondary's master — users can orphan after failover.
- Script logins with matching SIDs onto every replica (sp_help_revlogin), or use AD groups / contained DBs.
- Fix orphans with ALTER USER ... WITH LOGIN.

Q99. Difference between DB_FAILOVER ON and OFF?

- ON (SQL 2016+): a database-level failure (suspect/offline) triggers a failover of the whole AG.
- OFF: only instance-level health issues trigger failover.
- With OFF, a single unhealthy database wouldn't cause a failover.

Q100. How do you monitor AGs proactively in production?

- Alert on send/redo queues and sync state from the DMVs.
- Watch the AlwaysOn_health XE session and alert on failover events.
- Track HADR_SYNC_COMMIT waits and dashboard RPO/RTO estimates.
- Baseline normal queue sizes so anomalies stand out.

Q101. Top best practices for a production AG deployment?

- Identical hardware/spec on all replicas; sync partners on low-latency links.
- Proper witness for odd quorum; apps connect only via the listener with MultiSubnetFailover=True + retry logic.

- Automate AG-aware backups; script logins/jobs to all replicas (or use contained AGs).
- Test failover regularly; monitor queues, waits, and RPO/RTO continuously.

Q102. How do you validate that failover actually works before relying on it?

- Schedule regular planned failover drills in a maintenance window.
- Confirm the listener follows, apps reconnect, jobs run on the new primary, backups/monitoring continue.
- Document observed RTO vs target — untested failover is unproven HA.

Q103. Recommended approach for very large databases (VLDBs) in an AG?

- Use manual backup/restore seeding over compressed backups instead of auto-seeding on a slow link.
- Place data/log/tempdb on fast storage; keep transactions small to limit redo bursts.
- Consider Distributed AGs to stage cross-region copies without hammering the primary.